

Deepak Mallampati

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Kent, OH - Open to EU relocation (work-permit sponsorship needed)

PROFESSIONAL SUMMARY

Applied AI engineer with end-to-end RAG and agentic-workflow delivery, FastAPI/Docker service packaging, and a track record of cutting post-deploy defects ~30% via load simulation and structured logging. Grafana-based observability and CI/CD discipline from an oil-and-gas ERP background (~25% incident recurrence reduction). Strong production reliability instincts applied to LLM/ML services - the gap between 'demo works' and 'production stays up at scale' is where I do my best work. Open to EU relocation with work-permit sponsorship.

CORE COMPETENCIES

Production Reliability • FastAPI / Docker • Grafana Observability • CI/CD (Jenkins) • RAG / Agentic Workflows • LLM Evaluation • Structured Logging • PyTorch / Python • SQL (T-SQL, PostgreSQL)

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - USA (Healthcare Technology, AI Consulting, SaaS)

- Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking + transformer embeddings improved contextual retrieval precision by ~**35%** and cut irrelevant retrieval >**30%**; grounded response alignment exceeded **90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; raised agent response stability ~**25%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement, and support prioritization; improved recall **15-20%** on high-risk categories while holding precision >**90%** via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI/Flask** services, containerized with **Docker**, with structured logging and load simulation; reduced post-deploy defects ~**30%**.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, support ticket logs; raised dataset reliability above **98%** and cut downstream model instability ~**40%**.
- Maintained HIPAA-conscious data governance: de-identification, lineage docs, evaluation audit trails, stakeholder-facing limitation documentation.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India (Oil & Gas / ERP)

- Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform; reduced query execution time ~**20%** and data retrieval latency ~**25%**.
- Built **CI/CD pipelines with Jenkins** for schema and stored-procedure deployments; cut deployment errors >**30%** and release cycle ~**35-40%**.
- Designed performance dashboards via SQL Server DMVs and **Grafana** for long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~**25%**.
- Tuned database maintenance (index rebuilds, statistics updates, partition-aware indexing); improved reconciliation job runtime ~**15%**, reduced deadlocks ~**15%**.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Hyderabad, India (Nonprofit / Video Analytics / Applied NLP)

- Built a **transformer-based video summarization & highlight system** over 5,000+ recorded sessions; hierarchical long-context summarization + timestamp-aligned clip extraction reduced manual review time **60-70%**.
- Achieved ~**85% alignment** between AI-selected highlights and human-curated key moments.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); cut hallucinated outputs ~**30%** and raised contextual accuracy >**85%** in controlled tests.
- Deployed as a **Flask**-based API with a lightweight UI for non-technical staff.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app interpreting dreams and synthesizing dreamscapes via coordinated multimodal APIs (Perplexity Sonar for interpretation, Replicate image models for 16:9 visuals). Introduced **intermediate structured prompt transformation layers** - improved contextual alignment ~30% and first-pass image success rate ~25-30% over naïve direct-prompt orchestration.

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped, Chrome Web Store

Full-stack browser extension (Manifest V3, content scripts, service workers, captureVisibleTab). Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-section capture pipeline handles tall webtoon panels via viewport-slice screenshots with coordinate remapping and dedup. Seven-day translation cache, per-domain selector configs for 29 manga/webtoon sites, privacy policy, narrowed host permissions.

Video Summarization & Highlight Generation System

Transformer-based hierarchical abstractive summarization mapped back to source timestamps for automated clip extraction. **60-70% reduction** in manual review time, ~85% **highlight alignment** with human curation, Flask-based API for non-technical users.

Driver Drowsiness Detection System - Real-Time YOLOv8 Fatigue Monitoring

Replaced a two-stage CNN pipeline with a **unified YOLOv8** detection-and-classification model; reduced inference latency ~30%. Sliding-window confidence aggregation to suppress blink-driven false positives (~25% **reduction**), adaptive frame skipping, NMS tuning for stable real-time operation via OpenCV.

EDUCATION

M.S., Kent State University - Ohio, USA

Aug 2023 - May 2025

WORK AUTHORIZATION

F-1 OPT - authorized to work in the United States; future sponsorship required.

SKILLS

ML Infrastructure: Docker, FastAPI, Flask, Jenkins, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging, structured logging, load simulation

ML Libraries & Training: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, Diffusers, walk-forward validation, threshold calibration

LLM & GenAI: RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, embeddings, vector search, guardrails

Languages: Python, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, NetworkX

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision